



Name: _____

Teacher: _____

Year 11
Mathematics Extension 1
Assessment Task 3
2017

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour 30 minutes
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 13, show relevant mathematical reasoning and/or calculations

Total marks – 55

Section I

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II

45 marks

- Attempt Questions 11 – 13
- Allow about 75 minutes for this section

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Section I

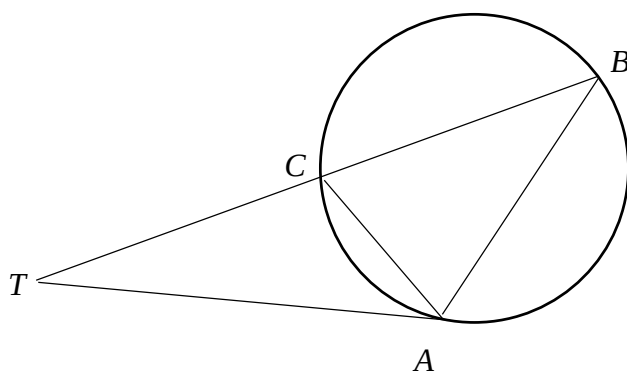
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Which of the following quadrilaterals are always concyclic?
- (A) Parallelogram and Rhombus
 - (B) Trapezium and Rectangle
 - (C) Square and Rhombus
 - (D) Square and Rectangle
- 2 Line TA is the tangent to the circle at A , and TB is a secant meeting the circle at B and C .



Which of the following is correct?

- (A) $\angle TAC = \angle ABC$ (Alternate Segment Theorem)
- (B) $\angle TAC = \angle ACB$ (Alternate Segment Theorem)
- (C) $\angle ACB = \angle ABC$ (Angles in the same segment are equal)
- (D) $\angle TAC = 2\angle ABC$ (Angles at the centre are twice the angle at the circumference, subtended by the same arc)

- 3 12 students are studying Spanish and another 15 students are studying French. How many ways can a student council be formed that comprises of 3 students who study Spanish and 4 who study French?

(A) ${}^{12}P_3 + {}^{15}P_4$

(B) ${}^{12}P_3 \times {}^{15}P_4$

(C) ${}^{12}C_3 + {}^{15}C_4$

(D) ${}^{12}C_3 \times {}^{15}C_4$

- 4 The point $P(x, y)$ divides the line joining the points $A(5, -2)$ and $B(-1, -5)$ **externally** in the ratio 1:2.

What are the coordinates of P ?

(A) $(3, -1)$

(B) $(1, -2)$

(C) $(-7, -8)$

(D) $(11, 1)$

- 5 Which of the following is equal to $\sec \theta$ in terms of t where $t = \tan \frac{\theta}{2}$.

(A) $\frac{2t}{1-t^2}$

(B) $\frac{1-t^2}{2t}$

(C) $\frac{1+t^2}{1-t^2}$

(D) $\frac{1+t^2}{2t}$

6 Find the exact value of $\sin 75^\circ$.

(A) $\frac{1+\sqrt{3}}{2\sqrt{2}}$

(B) $\frac{1-\sqrt{3}}{2\sqrt{2}}$

(C) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

(D) $\cos 25^\circ$

7 Solve $\sin^2 x = \frac{3}{4}$ for $0^\circ \leq x \leq 360^\circ$.

(A) $x = 30^\circ, 150^\circ$

(B) $x = 60^\circ, 120^\circ$

(C) $x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$

(D) $x = 60^\circ, 120^\circ, 240^\circ, 300^\circ$

8 Find $\lim_{x \rightarrow \infty} \frac{2x^3}{4x^3 - 1}$

(A) $\frac{1}{2}$

(B) 2

(C) 0

(D) ∞

9 Four couples sit at a round table. In how many ways can they be arranged if each couple sits together?

- (A) 6
- (B) 8
- (C) 96
- (D) 630

10 Which of the following equates to $\frac{1 - \cos 2x}{\sin 2x}$?

- (A) $\cot 2x$
- (B) $\cot x$
- (C) $\tan x$
- (D) $\cos x$

Section II

45 marks

Attempt Questions 11–13

Allow about 75 minutes for this section

Answer each question in a **separate writing booklet**. Extra writing booklets are available.

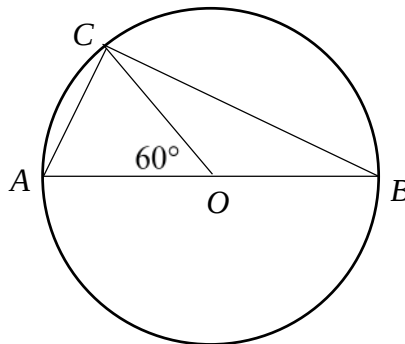
In Questions 11–13, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

(a) Solve $\frac{4}{x-3} \geq -2$

3

- (b) In the diagram below, AB is a diameter, passing through the centre O and $\angle AOC = 60^\circ$



- (i) Find the size of $\angle ABC$, give a reason for your answer

2

- (ii) Find the size of $\angle ACB$, give a reason for your answer

2

Question 11 continues on page 8

Question 11 Continued

- (c) In how many ways can the letters of the word EXTENSION be arranged in a line? **2**
- (d) A class consists of 13 students. At the end of a lesson, 1 student is to be chosen to clean the board and another student to put up the chairs. **1**
- In how many different ways can the two student be chosen?
- (e) A country's numberplates consist of 2 upper case letters, 3 digits and then another upper case letter such as PQ707E.
- (i) How many different numberplates are possible given letters and digits may be used more than once? **1**
- (ii) What is the probability that a numberplate produced at random will be PQ707E? **1**
- (f) Solve $\sin 2x = \sin x$, given $0 \leq x \leq 360$. **3**

End of Question 11

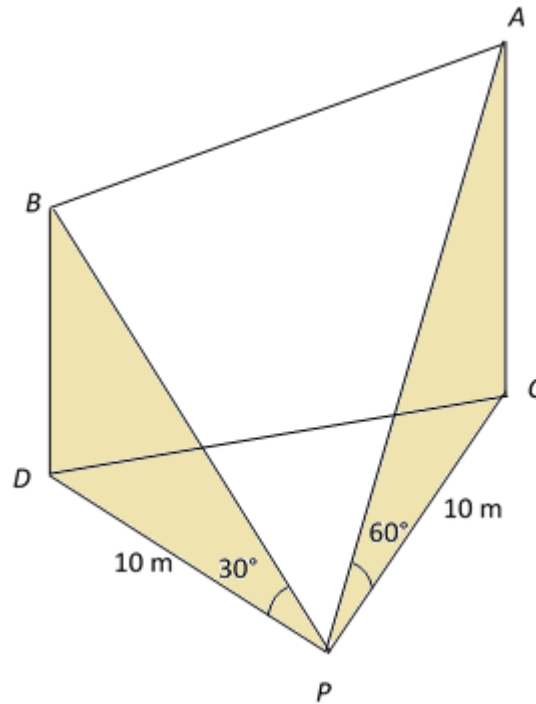
Question 12 (15 marks)

- (a) The graphs of the curve $f(x) = x^3$ and the line $y = -x + 2$ intersect at $(1,1)$.
- (i) Find the equation of the tangent to the curve $f(x) = x^3$ at $(1,1)$. **3**
- (ii) Find the value of the acute angle between the given line and the tangent to the curve at the point of intersection. **2**
- Give your answer correct to the nearest degree.
- (b) An extension Maths class has 12 students. 5 students are chosen at random to represent the class in an equation solving competition. **2**
- What is the probability that the chosen students are the first 5 on the roll alphabetically?
- (c) Find the points on the curve $f(x) = x^3 - 3x^2 + 4x$ where the tangent is parallel to the line $y = 4x - 5$. **3**
- (d) (i) Express $\cos \theta + \sqrt{3} \sin \theta$ in the form $r \sin(\theta + \alpha)$ where α is acute. **2**
- (ii) Hence or otherwise, solve $\cos x + \sqrt{3} \sin x = 1$, given $0^\circ \leq x \leq 360^\circ$. **3**

End of Question 12

Question 13 (15 marks)

- (a) Two drones, drone A and drone B , are hovering above a flat field. From a point P on the field, the angle of elevation of drone A is 60° and the angle of elevation of drone B is 30° on the field such that PC and PD are each 10 metres. Drone A is directly above the point C on the field and drone B is directly above the point D .



- (i) The bearing of drone A from P is 040° and the bearing of drone B from P is 310° . Show that $\angle CPD = 90^\circ$ and hence $CD = 10\sqrt{2}$ m. **2**
- (ii) Show that $3BD = AC$ **2**
- (iii) Using (i) and (ii), or otherwise, find the angle of elevation of drone A from drone B . Give your answer correct to the nearest degree **2**

Question 13 continues on page 11

Question 13 Continued

- (b) By first factorising, sketch the graph of $y = \frac{x^2 - x - 6}{x - 3}$.

2

Use about half a page for your graph.

- (c) Consider the function $y = \frac{x^2 + 2}{x}$

(i) Show that $\frac{x^2 + 2}{x} = x + \frac{2}{x}$.

1

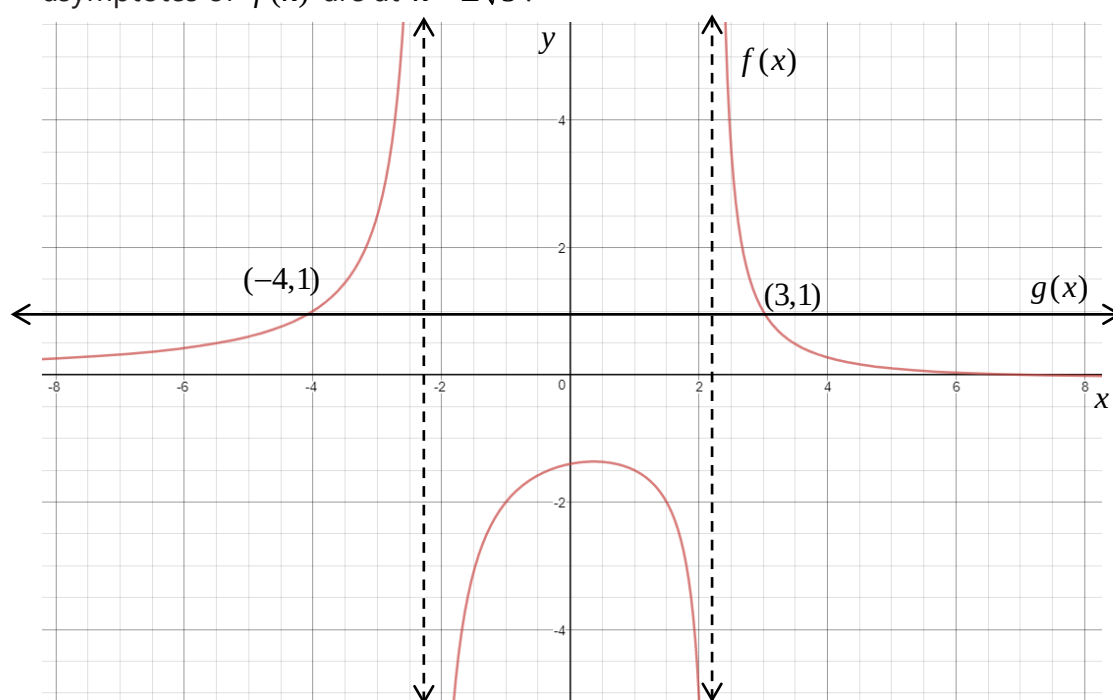
(ii) State any vertical, horizontal or oblique asymptotes of the function.

2

(iii) Without using calculus, graph the function, showing all asymptotes.
Use about half a page.

2

- (d) The functions $f(x) = \frac{7-x}{x^2-5}$ and $y=1$ are graphed below. The vertical asymptotes of $f(x)$ are at $x = \pm\sqrt{5}$.



Using the graph, give all solutions to the inequation $\frac{7-x}{x^2-5} \leq 1$

2

End of Paper

Mathematics Extension 1

Section I Multiple-Choice Answer Sheet

1 A ☐ B ☐ C ☐ D ☐2 A ☐ B ☐ C ☐ D ☐3 A ☐ B ☐ C ☐ D ☐4 A ☐ B ☐ C ☐ D ☐5 A ☐ B ☐ C ☐ D ☐6 A ☐ B ☐ C ☐ D ☐7 A ☐ B ☐ C ☐ D ☐8 A ☐ B ☐ C ☐ D ☐9 A ☐ B ☐ C ☐ D ☐10 A ☐ B ☐ C ☐ D ☐



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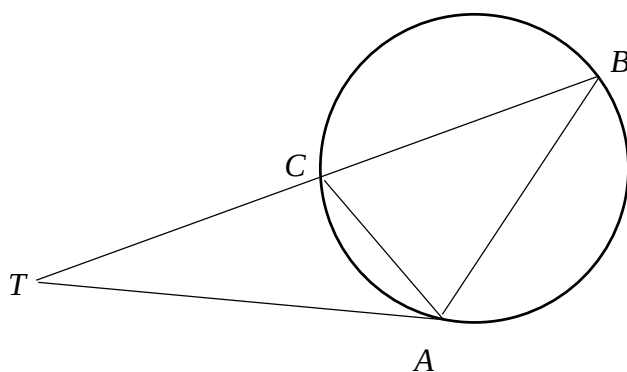
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(D) $\frac{1+t^2}{2t}$

6 Find the exact value of $\sin 75^\circ$.

(A) $\frac{1+\sqrt{3}}{2\sqrt{2}}$

(B) $\frac{1-\sqrt{3}}{2\sqrt{2}}$

(C) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

(D) $\cos(25)$

7 Solve $\sin^2 x = \frac{3}{4}$ for $0^\circ \leq x \leq 360^\circ$

(A) $x = 30^\circ, 150^\circ$

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Question 11 (15 marks)

(a) Solve $\frac{4}{x-3} \geq -2$

3

$$\frac{4}{x-3} \geq -2$$

$$x \neq 3$$

$$4 = -2(x-3)$$

$$4 = -2x + 6$$

$$2x = 2$$

$$x = 1$$

$$x \leq 1, x > 3$$

$$\frac{4}{x-3} \geq -2$$

$$4(x-3) \geq -2(x-3)^2$$

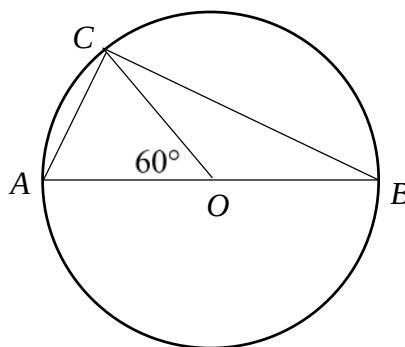
$$\text{or } 4(x-3) + 2(x-3)^2 \geq 0$$

$$2(x-3)[2 + (x-3)] \geq 0$$

$$2(x-3)(x-1) \geq 0$$

$$x \leq 1, x > 3$$

- (b) In the diagram below, AB is a diameter, passing through the centre O and $\angle AOC = 60^\circ$



- (i) Find the size of $\angle ABC$, give a reason for your answer

2

$\angle ABC = 30^\circ$ (Angle at the centre is twice the angle at the circumference subtended by the same chord/arc)

- (ii) Find the size of $\angle ACB$, give a reason for your answer

2

$\angle ACB = 90^\circ$ (Angle in a semicircle)

Question 11 Continued

- (c) In how many ways can the letters of the word EXTENSION be arranged in a line? 2

$$\frac{9!}{2!2!} = 90720$$

- (d) A class consists of 13 students. At the end of a lesson, 1 student is to be chosen to clean the board and another student to put up the chairs. 1

In how many different ways can the two student be chosen?

$$13 \times 12 = 156 \quad \text{or} \quad {}^{13}P_2 = 156$$

- (e) A country's numberplates consist of 2 upper case letters, 3 digits and then another upper case letter such as PQ707E.

- (i) How many different numberplates are possible given letters and digits may be used more than once? 1

$$26 \times 26 \times 10 \times 10 \times 10 \times 26 = 17576000$$

- (ii) What is the probability that a numberplate produced at random will be PQ707E? 1

$$\frac{1}{17576000}$$

- (f) Solve $\sin 2x = \sin x$, given $0 \leq x \leq 360$. 3

$$\begin{aligned} \sin 2x &= \sin x \\ 2 \sin x \cos x &= \sin x \\ 2 \sin x \cos x - \sin x &= 0 \\ \sin x [2 \cos x - 1] &= 0 \\ \sin x &= 0 \\ x &= 0^\circ, 180^\circ, 360^\circ \\ \cos x &= \frac{1}{2} \\ x &= 60^\circ, 300^\circ \end{aligned}$$

End of Question 11

Question 12 (15 marks)

- (a) The graphs of the curve $f(x) = x^3$ and the line $y = -x + 2$ intersect at $(1,1)$.

- (i) Find the equation of the tangent to the curve $f(x) = x^3$ at $(1,1)$. **3**

$f'(x) = 3x^2$	$y - y_1 = m(x - x_1)$
$f'(1) = 3(1)^2$	$y - 1 = 3(x - 1)$
$= 3$	$y = 3x - 2$
	$3x - y - 2 = 0$

- (ii) Find the value of the acute angle between the given line and the tangent to the curve at the point of intersection. **2**

Give your answer correct to the nearest degree.

$\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $
$\tan \theta = \left \frac{3 - (-1)}{1 + 3(-1)} \right $
$\tan \theta = 2$
$\theta = 63^\circ$

- (b) An extension Maths class has 12 students. 5 students are chosen at random to represent the class in an equation solving competition. **2**

What is the probability that the chosen students are the first 5 on the roll alphabetically?

$$\frac{1}{{}^{12}C_5} = \frac{1}{792}$$

- (c) Find the points on the curve $f(x) = x^3 - 3x^2 + 4x$ where the tangent is parallel to the line $y = 4x - 5$. **3**

$f'(x) = 3x^2 - 6x + 4$	$f(0) = (0)^3 - 3(0)^2 + 4(0)$
$3x^2 - 6x + 4 = 4$	$= 0$
$3x^2 - 6x = 0$	$(0, 0)$
$3x(x - 2) = 0$	$f(2) = (2)^3 - 3(2)^2 + 4(2)$
$x = 0, 2$	$= 4$
	$(2, 4)$

- (d) (i) Express $\cos \theta + \sqrt{3} \sin \theta$ in the form $r \sin(\theta + \alpha)$ where α is acute. **2**

$$\begin{aligned} r \sin(\theta + \alpha) &= r \sin \theta \cos \alpha + r \sin \alpha \cos \theta \\ \cos \theta + \sqrt{3} \sin \theta &= \sqrt{3} \sin \theta + \cos \theta \\ r \cos \alpha &= \sqrt{3} \\ r \sin \alpha &= 1 \\ r &= \sqrt{\sqrt{3}^2 + 1^2} \\ &= 2 \\ \tan \alpha &= \frac{1}{\sqrt{3}} \\ \alpha &= 30^\circ \\ \cos \theta + \sqrt{3} \sin \theta &= 2 \sin(\theta + 30)^\circ \end{aligned}$$

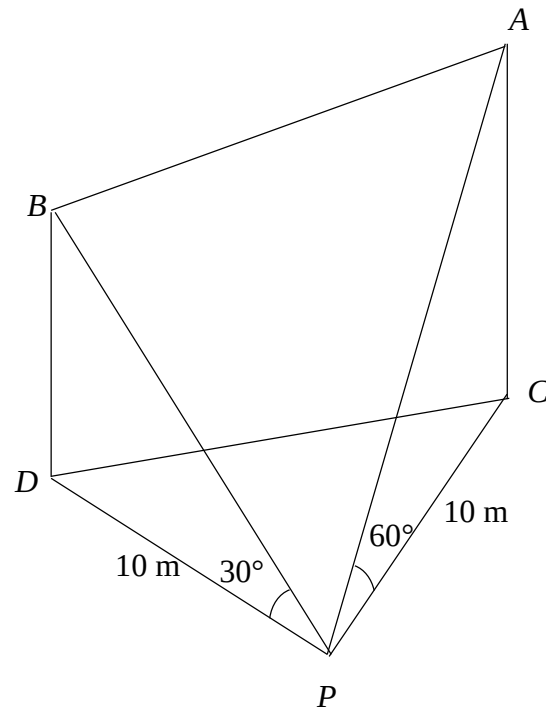
- (ii) Hence or otherwise, solve $\cos x + \sqrt{3} \sin x = 1$, given $0^\circ \leq x \leq 360^\circ$. **3**

$$\begin{aligned} 2 \sin(\theta + 30)^\circ &= 1 \\ \sin(\theta + 30)^\circ &= \frac{1}{2} \\ (\theta + 30)^\circ &= 30^\circ, 150^\circ \\ \theta &= 0^\circ, 120^\circ \end{aligned}$$

End of Question 12

Question 13 (15 marks)

- (a) Two drones, drone A and drone B , are hovering above a flat field. From a point P on the field, the angle of elevation of drone A is 60° and the angle of elevation of drone B is 30° on the field such that PC and PD are each 10 metres. Drone A is directly above the point C on the field and drone B is directly above the point D .



- (i) The bearing of drone A from P is 040° and the bearing of drone B from P is 310° . Show that $\angle CPD = 90^\circ$ and hence $CD = 10\sqrt{2}$ m. **2**

(Let N be the point due North of P)

$\begin{aligned}\angle CPD &= \angle DPN + \angle NPC \\ &= (360 - 310)^\circ + 40^\circ \\ &= 90^\circ\end{aligned}$	$\begin{aligned}CD^2 &= PD^2 + PC^2 \\ &= 10^2 + 10^2 \\ &= 200 \\ CD &= \sqrt{200} \\ &= 10\sqrt{2} \text{ m}\end{aligned}$
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(ii) Show that $3BD = AC$

2

$$\begin{array}{lll} \tan 30 = \frac{BD}{PD} & \tan 60 = \frac{AC}{PC} & \frac{AC}{3} = \frac{10\sqrt{3}}{3} \\ \frac{1}{\sqrt{3}} = \frac{BD}{10} & \sqrt{3} = \frac{AC}{10} & = \frac{10}{\sqrt{3}} \\ BD = \frac{10}{\sqrt{3}} & AC = 10\sqrt{3} & = BD \end{array}$$

(iii) Using (i) and (ii), or otherwise, find the angle of elevation of drone A from drone B. Give your answer correct to the nearest degree

2

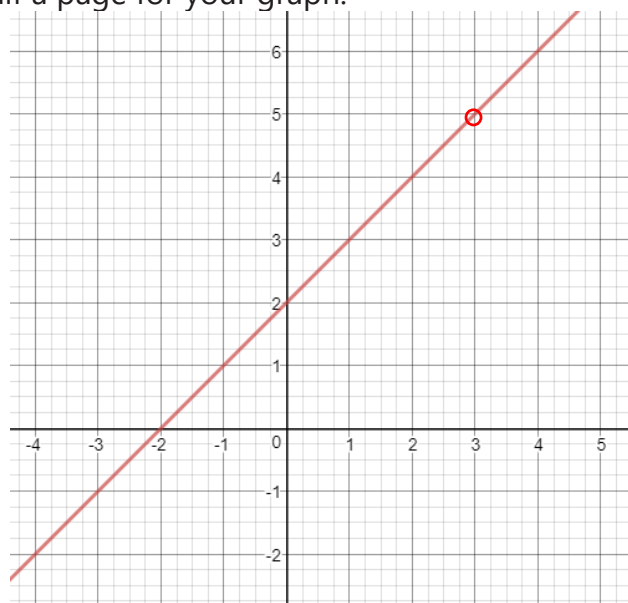
BD is y metres, AC is 3y metres, so AC is 2y units taller than BD

$$\begin{array}{l} \tan \theta = \frac{\frac{2}{3} \times \sqrt{3}x}{\sqrt{2}x} \\ \tan \theta = \frac{2\sqrt{3}}{3\sqrt{2}} \\ \theta = 39^\circ \end{array}$$

(b) By first factorising, sketch the graph of $y = \frac{x^2 - x - 6}{x - 3}$.

2

Use about half a page for your graph.



p

(c) Consider the function $y = \frac{x^2 + 2}{x}$

(i) Show that $\frac{x^2 + 2}{x} = x + \frac{2}{x}$.

1

$$\begin{aligned}\frac{x^2 + 2}{x} &= \frac{x^2}{x} + \frac{2}{x} \\ &= x + \frac{2}{x}\end{aligned}$$

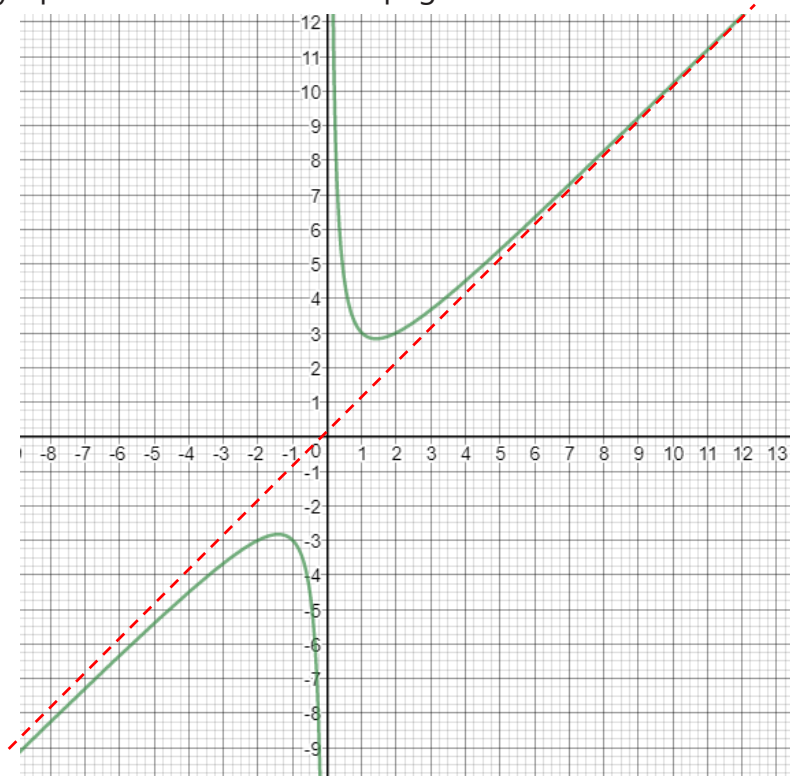
(ii) State any vertical, horizontal or oblique asymptotes of the function.

2

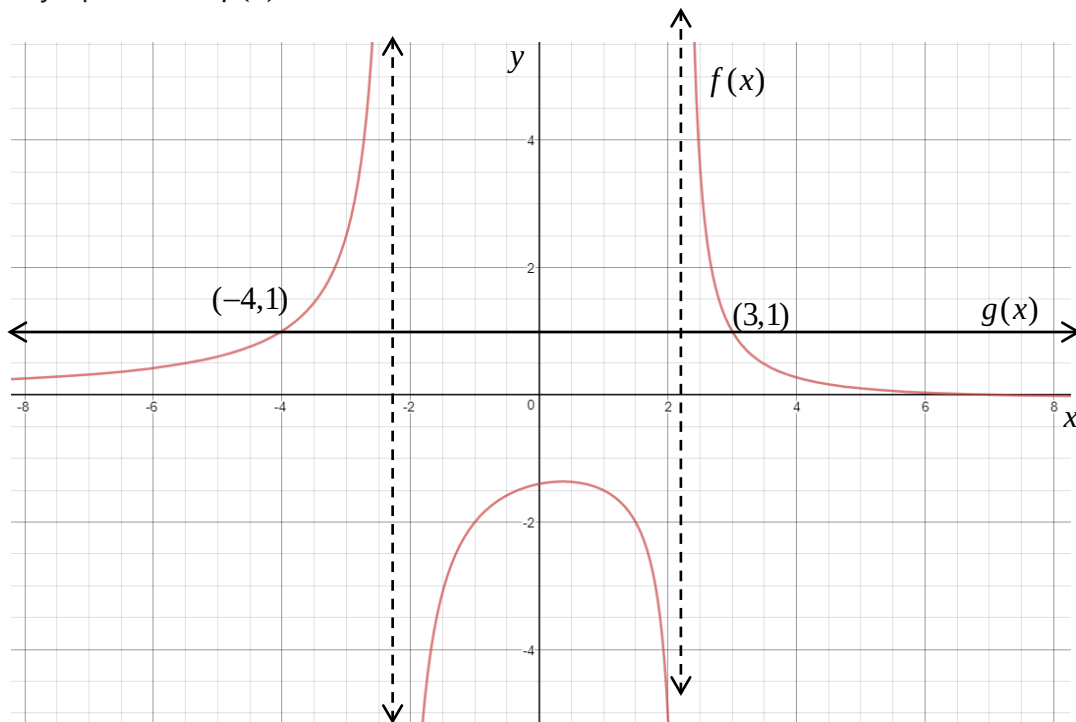
Vertical at $x = 0$, Oblique at $y = x$

(iii) Without using calculus, graph the function, showing all asymptotes. Use about half a page.

2



- (d) The functions $f(x) = \frac{7-x}{x^2-5}$ and $g(x) = 1$ are graphed below. The vertical asymptotes of $f(x)$ are at $x = \pm\sqrt{5}$.



Using the graph, give all solutions to the inequation $\frac{7-x}{x^2-5} \leq 1$.

2

$$x \leq -4, \quad -\sqrt{5} < x < \sqrt{5}, \quad x \geq 3$$

End of Paper

Mathematics Extension 1

Section I Multiple-Choice Answer Sheet

- | | | | | |
|----|------------------------------------|-------------------------|------------------------------------|------------------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☒ |
| 2 | A ☒ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☒ |
| 4 | A ☐ | B ☐ | C ☐ | D ☒ |
| 5 | A ☐ | B ☐ | C ☒ | D ☐ |
| 6 | A ☒ | B ☐ | C ☐ | D ☐ |
| 7 | A ☐ | B ☐ | C ☐ | D ☒ |
| 8 | A ☒ | B ☐ | C ☐ | D ☐ |
| 9 | A ☐ | B ☐ | C ☒ | D ☐ |
| 10 | A ☐ | B ☐ | C ☒ | D ☐ |